Viewpoint Invariance Is a Theorem, Not a Fit: An $SE(3)$-Equivariant Recurrence for Rehabilitation Exercise Assessment

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Abstract

A clinical screening instrument must return the same patient the same score from a different room. We show that for skeleton-based rehabilitation assessment this per-prediction guarantee is available by construction: an $SE(3)$-equivariant recurrent model whose read-out is invariant to camera viewpoint as a theorem*, certified to the fp64 roundoff floor ($\sim 10^{-16}$) over Haar-random $SO(3)$ — of which any camera relocation, including the $45°$ NTU RGB+D Cross-View displacement, is one instance. On clean frontal data no model separates from another — ours, a point-cloud-transformer baseline, and a hand-crafted-invariant GRU tie within a 0.33 MAD nondeterminism floor — so the benchmark's aggregate metric is saturated and the decisive axis lies elsewhere: the augmented baseline matches us in aggregate accuracy while still swinging an individual patient's score by up to 20.1 MAD (of 50) under camera rotation, against our 9×10^{-6}. We isolate what the steerable encoder buys with a hostile stress suite: under sensor-node failure the graph-pooled equivariant model degrades gracefully where a hand-crafted-invariant control collapses through the mean-predictor floor, and a lighter $E(n)$-equivariant encoder behind the identical invariant cut is measurably more brittle. We replicate the viewpoint and node-failure properties on a second corpus (REHAB24-6), and measure what the theorem costs on the edge: an int8 activation-quantization cliff of only 0.05 MAD — one to two orders of magnitude below MCID thresholds on comparable rehabilitation scales (Sec. 4) — and a causal streaming variant at 112 fps (GPU, batch 1). We report our negative results — a non-working continuous-time variant and a null irregular-sampling regime — with equal prominence.*

1. Introduction

Automated assessment of rehabilitation exercises from skeletons promises objective remote monitoring [1]. The dominant regime, however, trains and tests on clean frontal recordings from one sensor and ranks models by a single aggregate error against a clinician's score [2–5]. It rewards capacity and penalizes nothing a deployed device meets: a camera against a different wall, a tracking node that freezes, a dropped frame.

This paper argues that the *regime*, not the architectures, is the problem. On clean frontal KIMORE no architecture separates from another: our model, the point-cloud transformer, and a GRU on six hand-crafted invariant features are statistically indistinguishable — the benchmark's aggregate metric is saturated. We can say so precisely because we first measured what a difference must exceed to be one: training here is not bit-reproducible (nondeterministic cuDNN and scatter-add kernels), and repeated fits of an *identical* configuration differ by ≈ 0.33 MAD — larger than every clean-accuracy gap between the leading models. A single-run comparison on this benchmark reports its own seed. Worse, auditing the commonly used single-exercise slice, optimistic epoch selection is worth $+1.2 \pm 0.4$ MAD (18.6% over three seeds), and the honest number carries no reliable subject-level signal: the paired subject-cluster bootstrap gives $\Delta = -0.44$ (per-seed $[-1.07, +0.20]$), clearing zero in only one of three seeds. We therefore pool the five exercises, keep splits subject-disjoint, and gate every claim on clearing a per-exercise mean-predictor floor — a model that ignores its input has zero degradation and would win every robustness plot.

With clean accuracy neutralized, the question becomes what a structural guarantee buys. We formulate assessment as an *invariant read-out of an equivariant representation*: a steerable message-passing encoder produces per-joint features in $O(3)$ irreducible representations, an explicit projection onto a generating set of invariants discharges the symmetry, and a recurrence consumes the invariant code with the sensor's actual inter-arrival times. Invariance of the score under any rigid camera motion is then a *theorem* (Proposition 1), true for arbitrary weights, not a behaviour induced by augmentation. The theorem itself is standard — a function built from scalar contractions of Wigner-D-

covariant features is invariant by the representation theory the e3nn/tensor-field-network family already formalizes [10–12]; our contribution is not new invariance theory but the discipline of certifying it end to end (eight numerical gates surviving the real steerable layers, the adaptive solver, and int8 quantization) and the evaluation protocol — the floor, the audit, the per-sequence degradation curve — that makes a guarantee's practical value legible rather than assumed. The distinction bites: a rotation-augmented baseline has a flat *aggregate* accuracy curve across azimuth — by the field's metric it has solved viewpoint — yet its *per-sequence* degradation averages 3.0 MAD and reaches 20.1 in the tail, individual scores swinging up to two fifths of the fifty-point scale as the camera moves, the errors merely cancelling in the mean. Ours move by 9×10^{-6}. We therefore report degradation and accuracy side by side throughout.

The tie with the hand-crafted model leaves an obligation elegance cannot discharge: if a plain GRU on named features is exactly as accurate *and* as viewpoint-invariant, the steerable machinery is unjustified. We discharge it by measurement. When Kinect nodes freeze — the characteristic consumer-depth failure — the hand-crafted model fails outright: a *single* dead joint drives it from 6.31 to 9.78 MAD, through the floor, because its features *name* the joints they depend on; the graph encoder pools learned messages and dilutes the failure across 25 joints. We also treat the guarantee as a *systems* property: it survives to int8 (only activation-, not weight-rounding perturbs it) and converts to strictly causal streaming at 8.9 ms/frame. Finally we report, in full, the two places the structural story does not pay — a continuous-time variant that fails the floor, and a restored parity channel that is principled but unrewarded on healthy-form exercises.

Contributions. (i) A protocol audit that establishes what the benchmark can measure — an 18.6% epoch-selection inflation, a 0.33 MAD nondeterminism floor below which no accuracy gap is interpretable, and a gated pooled protocol. (ii) Exact viewpoint invariance, *certified* by eight numerical gates (E1–E8) over Haar-random $SO(3)$ rather than a single azimuth sweep, against a baseline that crosses the floor by $45°$. (iii) A measured justification for the steerable encoder — sensor-node failure — and, behind the identical invariant cut, evidence that lighter $E(n)$ equivariance is more node-fail brittle. (iv) A second-corpus replication (REHAB24-6) of the viewpoint and node-failure properties. (v) A co-designed deployment path: an int8 precision-budget audit and a causal streaming read-out with an honest time-to-first-score. We also report two negative results, one of them derived.

2. Related Work

Rehabilitation assessment and equivariant models. KIMORE [1] is the standard benchmark; reported results span ST-GCN variants with contrastive objectives [3], cross-modal augmentation [4], dual-stream graph networks [5], and point-cloud transformers [2], with Rehab-Pile [6] (aggregating KIMORE, UI-PRMD [7] and IRDS) the closest antecedent of our protocol discipline. Evaluation is uniformly in-distribution (same sensor, same placement, no simulated failure), and to our knowledge no prior work reports a mean-predictor floor, without which a degradation curve cannot be interpreted. We do not enter these papers' published numbers into Tab. 3 because none is protocol-comparable: Kuang et al. [5] report MAD 0.10–0.16 in the *same* 0–50 clinical-score units as our subject-disjoint 6.3–6.7 — lower by a factor of 39–67, with no rescaling in between. We recover those units from the published tables rather than assuming them (supplement): the reference implementation standardises its targets, so the reading is not self-evident, and getting it wrong would move the comparison by $8.5\times$. Their split is an explicit 8:1:1 *sample-level* stratified one, not a subject-disjoint one; Karlov et al. [3] report a Spearman correlation rather than MAD — we now report ρ too (Tab. 3), which makes the non-comparability legible rather than asserted: our pooled $\rho \approx 0.6$ sits below the 0.74–0.965 band that literature spans, on a protocol whose mean-predictor null we also state; Abedi et al. [4] report only a relative MAD reduction with no absolute baseline recoverable. We do not attribute these differences: we have neither their preprocessed data nor a subject-disjoint rerun of their pipelines, so we report the protocols as incomparable rather than diagnose their numbers, and treat the field's published values as measured under a different protocol rather than as an external accuracy ceiling to beat. Group-equivariant networks impose symmetry as a constraint on the hypothesis class [9]; the steerable family — tensor-field networks [10], $SE(3)$-Transformers [11], e3nn [12] — guarantees equivariance for arbitrary weights, while lighter constructions reach $E(n)$-equivariance without spherical harmonics [13] or via vector neurons [14]. We build on [12] but our claim is deliberately *not* about steerable machinery: a GRU on hand-crafted invariants attains the same accuracy and viewpoint guarantee at a third of the parameters, and we isolate the one axis on which they differ (Tab. 4).

Invariance by canonicalization, augmentation, and corruption robustness. A non-equivariant backbone can be made invariant by canonicalization or frame averaging [15], exact only in so far as the frame estimate is — a point estimate that turns discontinuous near covariance degeneracy, which we measure and confront directly (Sec. 4); our cut estimates no frame, being an explicit projection exact for arbitrary weights (Proposition 1). Training-time augmentation flattens *aggregate* error but makes per-prediction errors cancel, not vanish — expressible only if degradation is reported per sequence. For irregular sampling, the standard alternatives are resampling or continuous-time models [16–19]; we implement the Neural CDE variant (adaptive Dormand–Prince solver [20]) as a certified *control* that fails the floor. Occlusion and tracking failure are usually met with masking or attention [8]; we adopt the failure literally and use it as the discriminating experiment between two otherwise-indistinguishable models.

3. Method

A steerable encoder on the skeleton graph carries per-joint latents in $O(3)$ irreps (n_0 type-0 scalars s_j, n_1 type-1 vectors v_j); feeding *root-relative* coordinates makes translation invariance automatic. The encoder is *equivariant*; the score must be *invariant*.

The invariant cut. The transition is an explicit projection Π onto a generating set of invariants. The parity-even generators (scalars, vector norms, pairwise cosines) are complete for the full $O(3)$ — each is fixed by a reflection — so a cut built from them is blind to *trajectory handedness* (a clockwise and anticlockwise limb circle share every bone length, distance and speed). We complete it with parity-odd pseudo-scalar triple products over unit vectors:

$$\pi_j^{\text{even}} = \Big[\tanh s_j \,\|\, \log(1+\|v_j^c\|)_c \,\|\, \langle \hat{v}_j^c, \hat{v}_j^{c'} \rangle_{c<c'}\Big] \quad (1)$$

$$\pi_j^{\text{odd}} = \Big[\det[\hat{v}_j^a, \hat{v}_j^b, \hat{v}_j^c]_{a<b<c}\Big] \quad (2)$$

$$\pi_j = \big[\pi_j^{\text{even}} \,\|\, \pi_j^{\text{odd}}\big] \quad (3)$$

with $\hat{v} = v/\|v\|$, pooled over joints by mean$\oplus$max and concatenated with invariant bone lengths and a fixed set of anatomical signed volumes χ_τ (index-preserved, unpooled):

$$\Pi(h, \tilde{\boldsymbol{x}}) = \big[\,\text{mean}_j \pi_j \,\|\, \max_j \pi_j \,\|\, (d_{ij})_{\mathcal{E}} \,\|\, (\chi_\tau)_{\mathcal{T}}\,\big]. \quad (4)$$

At the deployed widths (n_0=32, n_1=8) the parity-even subtotal is 160-d and the parity-odd completion adds 123-d (283-d total; full specification in the supplement). A liveness gate zeroes any generator touching a failed joint, which makes the dead-node *invariance* certified by E1–E8 exact under a known failure mask; the degradation reported in Tab. 4 instead uses the deployed gate-*off* model with no failure signal supplied, so the graceful fall-off measured there is message-passing dilution alone, not the gate.

Proposition 1 (**$SO(3)$-invariance of the read-out**) *For an encoder that intertwines with the Wigner-D action on its latents, Π of Eq. (3)–(4) is invariant under $SO(3)$ for* arbitrary *weights; the predicted score is therefore invariant to any rigid motion of the camera. (Proof in the supplement.)*

Certificate. We audit the read-out with eight numerical gates (Tab. 1), all passed, each independently re-runnable and refutable (see Code availability): representation orthogonality $\le 10^{-12}$; violation at the fp64 roundoff floor ($\sim 10^{-16}$) with a *flat* log–log slope in rotation angle ($|\text{slope}| < 0.5$, against ≈ 1 for a genuine modelling error) and an fp32/fp64 violation ratio $> 10^8$ (excluding a small modelling violation masquerading as noise); translation and read-out invariance both $\sim 10^{-16}$ (fp64) after the parity-odd channels are admitted. The guarantee is over Haar-random proper rotations of the *full* $SO(3)$, of which a camera azimuth or a cross-view displacement is one instance. A continuous-time (Neural CDE) variant, certified to the same standard, *fails* the mean-predictor floor and is reported only as a control (supplement).

Table 1. **The viewpoint guarantee is falsifiable.** Eight numerical gates on the deployed read-out — E1–E5 that the invariance is *present*, E6–E8 that it is the *intended* one (chirality live, laterality preserved). All pass; any reviewer can rerun. Roundoff-floor quantities are regime-stable, not bit-reproducible (representative fp64 run); E6–E7 are model-dependent *modelling* quantities, not roundoff, and are reported for seed 0 of the released `certify_egru.json`; the parity/laterality gates report the even / chiral cut.

Gate	Criterion	Result
E1	rep. orthogonality $\|\rho^\top\rho - I\|_F$	$\le 10^{-12}$ ✓
E2	motion-channel invariance	verified ✓
E3	rotation-sweep violation / slope	$\sim 10^{-16}$, $\lvert\cdot\rvert < 0.5$ ✓
E4	fp32/fp64 violation ratio r	$> 10^8$ ✓
E5	translation invariance	$\sim 10^{-16}$ ✓
E6	parity (even / chiral)	0 / 4.4×10^{-2}
E7	laterality (even / chiral)	6.9 / 5.6×10^{-2}
E8	full-$SO(3)$, parity-odd active	$\sim 10^{-16}$ ✓

4. Results

All experiments are pooled, subject-disjoint, over 3 seeds, except the TCN, ST-GCN and Ridge viewpoint curves of Fig. 1, which are 5-fold cross-validated at a single seed. *Protocol audit* (Tab. 2): on the single-exercise slice most commonly reported, honest epoch selection removes an 18.6% inflation (3 seeds) and leaves no robust separation from a mean predictor — the paired subject-cluster bootstrap gives $\Delta = -0.44$ and clears zero in only 1/3 seeds. This is not subject leakage (each subject contributes one recording on that slice, so a random split is already subject-disjoint); the slice simply carries no robust subject-level signal, which is why we pool the five exercises and gate every experiment on a per-exercise mean-predictor floor.

The clean-accuracy metric is saturated. EGRU (6.73 ± 0.30), a GRU on hand-crafted invariants (Invariant-GRU, 6.31 ± 0.16), and the point-cloud transformer (PCT, 6.47 ± 0.20) are mutually indistinguishable: every gap is inside the 0.33 MAD seed floor (Tab. 3). This is a state-

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Table 2. Protocol audit, single-exercise slice (KIMORE Ex. 1; subject-disjoint 5-fold; 3 seeds). *Honest* selects the epoch on a held-out validation fold; *inflated* takes the per-epoch test minimum (the reference protocol). Δ is the paired subject-cluster bootstrap MAD(honest) − MAD(floor); an interval below 0 means the model beats the mean predictor. The subject-level signal is *seed-fragile* — it clears zero in only $1/3$ seeds.

Seed	Honest / Inflated	Selection bias	Δ vs. floor (95% CI)
0	6.90/5.45	+1.44 (20.9%)	+0.20 [−0.59, +1.03]
1	6.53/5.00	+1.54 (23.5%)	−0.46 [−1.60, +0.72]
2	5.84/5.17	+0.67 (11.4%)	**−1.07 [−1.98, −0.15]**
mean	6.42±0.44/5.21±0.19	+1.22 (18.6%)	−0.44

ment about the models *relative to each other*, not about whether any of them carries real subject-level signal: unlike the fragile single-exercise slice (Tab. 2, clears zero in only $1/3$ seeds), the pooled bootstrap Δ vs. floor is entirely below 0 for all three — PCT $-1.87\,[-2.72, -1.06]$, InvariantGRU $-2.03\,[-2.89, -1.23]$, EGRU $-1.60\,[-2.37, -0.88]$ — so the tie is a tie among models that each genuinely beat the mean predictor, not three models indistinguishable from noise. The tie survives the metric change: Spearman ρ separates the same models by at most 0.05 (Tab. 3), so saturation is a property of the benchmark, not of the error measure chosen to report it. ρ also needs its null stated — the mean predictor scores pooled $\rho = 0.17$ while ignoring its input entirely, and exactly 0 once the between-exercise score differences it exploits are removed. A benchmark on which three architectures of vastly different inductive bias cannot be told apart is not measuring architecture — which is the premise for evaluating on an axis that can. Restoring chirality ($O(3) \to SO(3)$) costs +0.11 MAD for EGRU — inside the floor — for 16.7% more parameters and the removal of a silent handedness assumption; we adopt the chiral model on that basis, not an accuracy one. The chiral read-out's wider seed spread (±0.30 vs ±0.08 for the parity-even model) reflects the added parity-odd capacity, not instability — every seed stays inside the floor.

One asymmetry against the baseline is worth closing: EGRU sees a one-hot exercise id and the floor is per-exercise, so pooled PCT alone was scored blind to the exercise. Re-training it with identical conditioning (3 seeds, both arms) changes nothing: clean 6.59 ± 0.07 vs. 6.47 ± 0.20, augmented 7.42 ± 0.22 vs. 7.47 ± 0.23 — inside the floor and opposite in sign — and the per-sequence swing is if anything *larger* (21.43 vs. 20.12 MAD worst sequence). Viewpoint fragility is not an artefact of withholding exercise identity; we report the unconditioned arm throughout.

Viewpoint invariance is a theorem, not a fit (Fig. 1). Across test-time camera azimuth the certified-invariant models are flat to machine precision (EGRU mean 9×10^{-6} MAD, worst single sequence 3×10^{-4}; InvariantGRU

Table 3. Clean-data accuracy (pooled, subject-disjoint, mean±std over 3 seeds). Every pairwise gap is inside the 0.33 MAD *fixed-configuration* floor (seed-to-seed std 0.48; the wider seed-variation table is in the supplement). Δ vs. floor is the SAME paired subject-cluster bootstrap statistic used in Tab. 2 (n=77 held-out subjects, pooled over 3 seeds × 5 folds, checkpoints re-evaluated with no retraining — see `pooled_bootstrap_eval.py`): every model's interval is entirely below 0, so the tie *among* models coexists with each one genuinely beating the floor. Spearman ρ is reported per seed on that seed's complete out-of-fold vector (never per fold), pooled over the five exercises and again *within* each exercise and averaged — the latter being the like-for-like comparison to a literature of single-exercise models. The floor row is why both are given: a predictor that ignores its input still attains pooled $\rho = 0.17$ purely from between-exercise score differences, and exactly 0 within exercise. Every ρ above it should be read as an increment over that null, not as an absolute.

Model	Group	MAD ↓	Params	Spearman ρ ↑ pooled	Spearman ρ ↑ within-ex.	Δ vs. floor (95% CI)
PCT (baseline)	—	$\mathbf{6.47 \pm 0.20}$	4.91M	0.60 ± 0.02	0.55 ± 0.02	−1.87 [−2.72, −1.06]
PCT + rotation aug.	—	7.47 ± 0.23	4.91M	—	—	—
InvariantGRU (hand-crafted)	$SO(3)$	$\mathbf{6.31 \pm 0.16}$	0.21M	0.59 ± 0.01	$\mathbf{0.56 \pm 0.01}$	−2.03 [−2.89, −1.23]
EGRU (ours)	$SO(3)$	$\mathbf{6.73 \pm 0.30}$	0.66M	0.56 ± 0.04	0.52 ± 0.04	−1.60 [−2.37, −0.88]
Mean-predictor floor	—	8.31 ± 0.05	—	0.17 ± 0.01	-0.07 ± 0.02	—

exactly 0; hand-crafted invariant features Ridge at the fp64 roundoff floor, 10^{-14} — both invariant by construction, so these two curves are a positive control, not independent evidence), while all non-equivariant baselines degrade catastrophically: PCT crosses the mean-predictor floor between 30° and 45° (mean per-sequence shift 9.33 MAD at its worst angle); TCN 13.17 MAD; ST-GCN 9.18 MAD. This pattern holds across three independently-designed non-equivariant architectures (temporal-conv, graph-conv, attention), indicating viewpoint fragility is a generic property of non-equivariance, not PCT-specific. Rotation augmentation flattens PCT's aggregate curve but leaves a *per-sequence* score shift averaging 3.01 MAD — and the average is the wrong statistic for a clinical claim: the 95th percentile is 8.65 MAD and the worst single held-out sequence moves 20.12, two fifths of the scale, purely because the camera moved. This is the difference between a guarantee and a fit. It is not an undertraining artefact: re-training the augmented arm at 3× and 9× the epoch budget leaves the shift flat (3.01/3.27/2.94 MAD mean at 60/180/540 epochs, worst sequence 20.1/21.4/19.9), so the residual swing is what augmentation converges to, not what it has yet to learn. The certificate is over Haar-random $SO(3)$, so the 45° NTU RGB+D 60 Cross-View camera displacement is one instance of the group it covers: invariance there is *inherited* from Proposition 1, not separately measured. What NTU contributes independently is generalisation, not invariance — trained *without* rotation augmentation on cameras 2–3 and tested on the displaced camera 1, the identical encoder reaches 82.98% Top-1 against a faithful ST-GCN

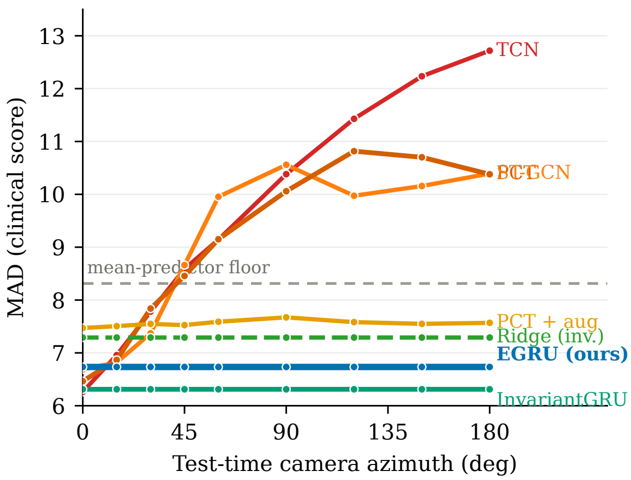

Figure 1. MAD vs. camera azimuth. Certified-invariant models (EGRU, InvariantGRU, hand-crafted Ridge) are flat by theorem; all non-equivariant baselines (PCT, TCN, ST-GCN) degrade 9–13 MAD at extreme angles, indicating viewpoint fragility is a generic property of non-equivariant architectures. Augmentation flattens PCT's aggregate but not its per-sequence swing (mean 3.01 MAD, worst sequence 20.12). Azimuth here is a rigid rotation of the ground-truth Kinect skeletons; it isolates the geometric effect of viewpoint and does not model per-view tracker degradation.

control's 80.38%.

Clinical interpretation of the MAD scale. No inter-/intra-rater reliability coefficient (ICC, κ) is reported for KIMORE's clinical Total Score [1] or for REHAB24-6's; a targeted search, including direct attempts to retrieve the Capecci et al. full text (both the publisher record and the indexed copy returned access-blocked to automated retrieval), could not surface one, and we do not fabricate one — we flag its absence as this paper's clearest open measurement gap. As external context, human inter-rater agreement on *movement-quality rating* tasks — the same construct KIMORE's clinicians perform, scoring how well an exercise was executed rather than a functional-recovery outcome — typically falls in a moderate-to-good range: the Cutting Movement Assessment Score reports inter-rater ICC 0.58–0.91, and a cross-diagnostic Movement Quality Score reports ICC 0.93. We read this as a defensible band on what a KIMORE-specific coefficient would plausibly show, not a substitute for one. As a secondary, magnitude-of-change cross-check from a different construct and scale, minimal clinically important differences (MCIDs) on comparable 0–66/0–100-point motor-*recovery* scales sit in the low single digits to low teens of points (Fugl-Meyer upper-extremity 4–12.4, lower-extremity 6, Stroke Rehabilitation Assessment of Movement 1.9–4.8 per subscale), putting the augmented baseline's per-sequence swing (mean 3.01, worst sequence 20.12, of 50) in the same order of magnitude as an established change-significance threshold on cognate instruments — at the tail, above it — and our own score shifts under rotation (9×10^{-6}), int8 and streaming (0.05 and lower, Sec. 5) orders of magnitude below it. A camera relocation that moves a patient's score by more than an MCID is a measurement the instrument cannot be trusted to make; that is the practical content of the guarantee. Both are plausibility bounds from adjacent instruments, not KIMORE-specific thresholds, and should be read as such.

Table 4. Sensor-node failure: MAD as k nodes freeze (mean over 3 seeds $\times$ 5 folds; identical hash-locked inputs; no retraining). Underline: past the floor (8.31). "+oracle": InvariantGRU given an oracle liveness mask (same checkpoint; features touching the dead joint zeroed at inference) — the measured fairness control.

Dead nodes k	EGRU (ours)	InvariantGRU	+oracle	PCT
1	7.10	9.78	9.39	6.76
4	8.41	15.96	15.25	7.51
8	10.50	18.37	16.33	8.73
MAD lost	+3.76	+12.05	+10.02	**+2.27**

What the steerable encoder buys: sensor-node failure (Tab. 4). Freezing k tracking nodes at their first-frame pose, the hand-crafted arm collapses through the floor at a *single* dead joint ($6.31 \to 9.78$) because its features name their joints; EGRU pools learned graph messages and loses only +3.76 MAD over 8 dead joints. Honestly, PCT is the most node-robust of the three (+2.27) — robustness to node failure is not a property of equivariance and we do not claim it is. The correct reading: *among models with an exact viewpoint guarantee*, only the graph-structured one survives a broken sensor up to $k{=}4$ dead nodes ($k{=}8$ crosses the floor for every model, EGRU included). Both models see identically corrupted joints with no imputation; we measured, rather than assumed, the fairness objection: granting the hand-crafted features an *oracle* liveness mask (told exactly which joint died, no retraining) closes only $\sim 25\%$ of the gap to EGRU ($+12.05 \to +10.02$ MAD lost; Tab. 4, right column) — the graph encoder needs no such signal and still loses far less, so the residual gap is architectural, not a missing-information artifact. The mechanism is invariant to the failure model (five operators incl. lag/burst/axis/lateral; supplement).

The mechanism is not an artifact of the freeze model. Under four further operators — stuck-at-lag, sporadic bursts, single-axis depth noise, and a left-limb-only variant — the ordering is invariant: the hand-crafted arm degrades most and the graph-pooled encoder least among the two exact-invariance models (Tab. 5). *Persistent* faults (freeze, lag) are far more destructive to naming features (+12.05, +8.88) than *transient* ones (burst +2.83, axis +1.93) — a length-masked temporal mean dilutes an intermittent spike

Table 5. Node failure under five operators (chiral models, 3 seeds $\times$ 5 folds, hash-locked). Each cell: MAD at k=1 / k=8 and total lost. The freeze/random row reproduces Tab. 4.

Failure operator	EGRU (ours)	InvariantGRU	PCT
freeze (random)	7.10/10.50 (+3.76)	9.78/18.37 (+12.05)	6.76/8.73 (+2.27)
freeze (left-only)	7.11/10.16 (+3.43)	10.81/15.04 (+8.73)	6.83/9.62 (+3.15)
stuck-at-lag	6.82/7.76 (+1.03)	9.02/15.19 (+8.88)	6.60/7.11 (+0.64)
sporadic burst	6.74/6.78 (+0.05)	6.69/9.14 (+2.83)	6.47/6.45 (−0.02)
axis/depth noise	6.77/7.20 (+0.47)	6.70/8.24 (+1.93)	6.48/6.50 (+0.03)

Table 6. Does each model clear the floor (8.31 MAD) under each stress? Last column: *mean* per-sequence score shift at the worst camera azimuth. The mean understates what a patient experiences: for PCT + rot the 95th percentile is 8.65 and the worst held-out sequence 20.12, against 2.8×10^{-5} and 3.0×10^{-4} for EGRU. EGNN shares the identical cut (viewpoint exact by construction). †Canon-PCA's near-zero is *empirical* — a frame estimate that happens to be stable on KIMORE — not a certificate; see text.

Model	Params	Clean	90°	2 dead	Mean degr.
EGRU (ours)	0.66M	6.73	6.73✓	7.55✓	$\mathbf{9\times10^{-6}}$
InvariantGRU	0.21M	6.31	6.31✓	11.64×	**0**
Lighter $E(n)$ (EGNN)	∼0.6M	6.88	6.88✓	8.44×	1.4×10^{-5}
Canon-PCA + PCT	4.91M	6.85	6.85✓	7.79✓	≈0†
PCT (baseline)	4.91M	6.47	10.13×	7.10✓	9.33
PCT + rot	4.91M	7.47	7.68✓	7.76✓	3.01

but not a permanently corrupted coordinate — and unilateral (left-only) failure does not break the parity channels (InvariantGRU +8.73, EGRU +3.43).

The whole thesis in one grid (Tab. 6). We do not claim to be the only survivor: rotation-augmented PCT also clears the floor under both a 90° move and two dead nodes. What separates the two survivors is the last column (Fig. 2a) — both flat in aggregate, but the augmented baseline's mean per-sequence degradation is 3.01 MAD (worst sequence 20.12) against our 9×10^{-6} (3.0×10^{-4} worst), at 7.4× more parameters. Behind the *identical* invariant cut, a lighter $E(n)$-equivariant (EGNN) encoder ties clean accuracy and viewpoint (exact by construction) but is measurably more node-fail brittle: tuned across depth, width, and coordinate-clamp over three seeds it still loses +6.4±1.3 MAD over eight dead joints (vs. EGRU's +3.76) and crosses the floor already at two dead joints (8.44 vs. EGRU's 7.55) — so the steerable encoder is the more robust of the two equivariant options here.

Canonicalization ties us offline — and the difference is the guarantee. The strongest objection to an equivariant model is that a non-equivariant backbone fed a canonical frame reaches the same place for less. It does, here: a per-frame PCA principal-axis frame feeding the transformer (Canon-PCA+PCT, Tab. 6) matches every offline axis — clean 6.85 MAD, 7.79 at two dead joints, and a viewpoint invariance that is empirically exact not merely for the azimuth subgroup a clinician sweeps but for arbitrary $SO(3)$ (worst per-sequence canonical-coordinate shift 3.7×10^{-11} over 32 random rotations). We report it as a first-class baseline rather than bury it. The distinction is not accuracy but *provenance of the invariance*: PCA canonicalization is a point estimate of a frame, exact only where the pose covariance is well-conditioned — and on KIMORE it often is not. 18% of frames sit within an eigen-gap of 0.05 of degeneracy, and the estimated frame rotates by more than 45° across 21% of frame transitions (sign flips to 180°; Fig. 2b); the downstream network learns to absorb this jitter on this corpus, but no theorem bounds it off distribution. Our cut estimates *no* frame: it is $SE(3)$-invariant for arbitrary weights by construction (Proposition 1), certified to roundoff over the full rotation group — so the *same* certificate covers, with no retraining, the genuinely displaced 45° camera of NTU RGB+D Cross-View. A frame estimator offers neither the worst-case bound nor that transfer. We do not claim an accuracy or on-distribution win over the handcrafted or canonicalized baselines here — the tie is total; we claim a *certified* worst-case that they structurally cannot offer, and we recommend the cheaper models where an off-distribution guarantee is not required.

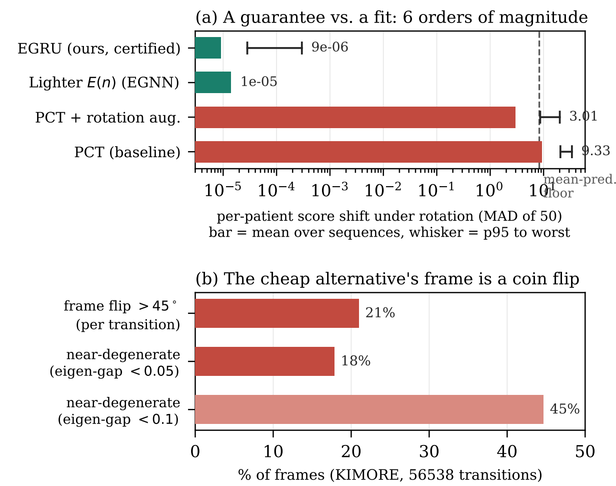

Figure 2. **The axis that separates the survivors.** (a) Per-patient score shift under camera rotation (log scale); bar is the mean over held-out sequences at the worst azimuth, whisker runs 95th percentile to worst sequence — the tail the mean conceals, and the reason we report both. EGNN has no measured tail and so carries no whisker. The certified models (EGRU, EGNN) hold a patient's score to $\sim 10^{-5}$ MAD, five to six orders of magnitude below the augmented and unaugmented transformers, whose worst sequences move 20.12 and 33.96 points of fifty. InvariantGRU is exactly 0 and Canon-PCA empirically ≈ 0, but unbounded off distribution. (b) Why the cheap canonicalization alternative is not a guarantee: its PCA frame flips by > 45° on 21% of frame transitions and sits within an eigen-gap of 0.05 of degeneracy on 18% of frames.

When the worst case is the deployment case. Concretely: a home tele-rehabilitation device is set up once by the patient, not a technician, and left running unattended — the camera sits wherever a phone stand or a laptop lid happens to land, and the consumer depth sensor is the component most likely to lose a joint (a hand behind the back, a node that freezes mid-repetition). Neither cheaper alternative is safe against *both* conditions at once. Canon-PCA+PCT's viewpoint guarantee is only empirical: Fig. 2b shows the PCA frame sits within an eigen-gap of 0.05 of degeneracy on 18% of frames — a fast or asymmetric repetition, a partially occluded limb — and a flipped frame there is silent, no flag, just a wrong score. InvariantGRU's viewpoint guarantee *is* exact, like ours, but its named-joint features make it the most node-fail-fragile model measured: one frozen tracking node drives it from 6.31 to 9.78 MAD, through the floor (Tab. 4), the same freeze mode a consumer sensor exhibits routinely. Uncontrolled placement and a tracking node that drops are exactly the two conditions a home device cannot rule out in advance, and EGRU is the only model measured that is provably safe against both at once. That is the niche this paper stakes out: not that InvariantGRU or Canon-PCA+PCT are worse choices in a supervised clinic with a fixed rig and a technician present, but that neither offers a bound that survives an unsupervised deployment on its own.

Second-corpus replication (REHAB24-6). To de-risk single-corpus reliance we replicate the structural story on REHAB24-6 (binary correct/incorrect repetitions; 1057 reps, 10 subjects; the identical 0.66M model as a BCE logit), an AUROC task over 3 seeds and subject-disjoint CV (Tab. 7). The accuracy tie holds — EGRU 0.738 ± 0.01 versus the transformer 0.707 ± 0.02 — and, decisively, the viewpoint theorem transfers: under a 90° camera rotation EGRU's AUROC is *unchanged* (logit drift $\leq 1.5 \times 10^{-4}$), while the transformer collapses to chance (0.52). Node failure degrades both gracefully. The guarantee is thus not a KIMORE artifact — though at 10 subjects this corpus is sized to replicate the *structural* properties (a certified invariance either transfers exactly or it does not) and should not be read as an independent accuracy benchmark; we do not draw a clean-accuracy conclusion from it beyond the tie already stated.

Per-family contribution of the cut. Zeroing one family of Eq. (3) at inference on the deployed chiral model (dimension preserved, no retraining) resolves the mechanism to family granularity (Tab. 8). Two families carry accuracy — the pairwise *cosines* (+1.83 MAD to remove) and the learned *triple products* (+2.75) — and removing the cosines is doubly harmful, inflating node-failure degradation the most (+7.29). The parity-odd families *split*: the learned triple products are relied upon while the fixed anatomical volumes are inert (+0.19), which reconciles rather than contradicts the chirality null — that null is a *retrain* comparison (parity-even families substitute during training), whereas this is an *inference* ablation on a model trained *with* the family.

Table 7. REHAB24-6 second-corpus replication (3 seeds, subject-disjoint CV). AUROC@90° is the same model evaluated under a 90° camera rotation: EGRU is invariant by theorem (logit drift $\leq 1.5 \times 10^{-4}$); the transformer falls to chance (drift 23.2). Node-fail: AUROC lost over 8 frozen nodes.

Model	AUROC	AUROC@90°	Node-fail lost
EGRU (ours)	$\mathbf{0.738 \pm 0.01}$	$\mathbf{0.738}$	$+0.14$
PCT (baseline)	0.707 ± 0.02	$0.52\times$	$+\mathbf{0.08}$

Table 8. Per-family ablation of the cut (each family zeroed at inference; 3 seeds $\times$ 5 folds). ΔAcc below the 0.33 floor is a tie; "nf lost" is worst MAD over 8 dead joints (full cut +3.76).

Family removed	Parity	Clean	ΔAcc	nf lost
— (full cut)	—	6.73	—	$+3.76$
cosines	even	8.56	$+\mathbf{1.83}$	$+\mathbf{7.29}$
triple products	odd	9.48	$+\mathbf{2.75}$	-0.16
bone lengths	even	7.40	$+0.67$	$+2.36$
scalars	even	6.98	$+0.25$	$+3.46$
vector norms	even	6.89	$+0.16$	$+3.51$
anat. volumes	odd	6.93	$+0.19$	$+3.43$

Two negative results, honestly — each a mapped boundary of where structure pays. (i) Native consumption of irregular arrivals does not help on KIMORE — the fixed-grid baseline is better at every drop level under a hash-locked budget — and a spectral census [21] predicts it: 97.7% of *positional* energy lies below the resampling corner (though 70% of *velocity* energy lies above it), so resampling preserves pose geometry while low-passing the derivative band — on KIMORE's healthy-form exercises, where the discriminative signal is positional, this acts as a denoiser; the boundary would move for tremor-dominated tasks. The mechanism is real (a 5 Hz tremor is recovered at r=1.000 through 70% drops where a resampled estimator decays to 0.785), so the claim is a scope boundary, not a defeat. (ii) Restoring chirality is principled but not rewarded on healthy-form exercises with no chiral pathology — the boundary at which a handedness prior stops paying. Both are stated with the corpus properties that would reward them (supplement).

5. Deployment Analysis

We treat the guarantee as a systems property (Tab. 9). *Precision budget:* the invariance floor climbs monotonically from fp64 to int8, and the int8 cliff is attributable *solely* to activation grid-rounding of the steerable channels —

Table 9. Deployment audit. Top: the *full* equivariance precision budget of the deployed pooled EGRU (INV floor = relative violation of the 160-d parity-even projection; score shift in MAD of 50). fp16 is 3.2× tighter than bf16 — equivariance pays in mantissa bits, which bf16 trades for exponent range — and the int8 cliff is *solely* activation grid-rounding (weight-only quant preserves the theorem). Bottom: causal streaming vs. bidirectional read-out.

Arithmetic	INV floor	Score shift	Note
fp64	5.75×10^{-16}	3.3×10^{-14}	exact (theorem)
fp32	2.68×10^{-7}	8.9×10^{-6}	= main viewpoint
fp16	2.62×10^{-3}	0.024	> bf16
bf16	8.34×10^{-3}	0.195	mantissa cost
int8, weights only	3.09×10^{-7}	1.2×10^{-5}	theorem holds
int8, activations only	3.38×10^{-2}	0.056	cliff (acts)
int8, weights+acts	$\mathbf{3.40 \times 10^{-2}}$	**0.051**	weight-quant. theorem-safe
Streaming: 8.9 ms/frame (112 fps); read-out cut unchanged; causal vs. bidir NTU X-View 79.8 vs. 83.0 (−3.2 pt).			

weight quantization preserves the theorem exactly (a quantized model is a different equivariant model) — while the reported clinical score still moves only 0.05 MAD, below the floor and one to two orders of magnitude below the comparator MCID band established in Sec. 4. *Streaming:* because invariance lives in a per-frame encoder and an invariant cut, the read-out converts to a strictly causal unidirectional form with a running-mean pool that emits a first score 8.9 ms after the first frame on an RTX-class GPU (batch 1; 112 fps, inside a 30 fps budget) — we report GPU latency as an upper bound on compute and leave edge-SoC profiling to deployment — reaching a usable prediction after ~ 55 frames (median), an honest time-to-first-*reliable*-score for real-time use. The invariant read-out cut is unchanged by the causal pooling, so Proposition 1 carries over verbatim; the cost is 3.2 points of NTU accuracy vs. the bidirectional model (NTU-60 X-View SOTA is $\sim 92\%$; our backbone is deliberately untuned — we measure the streaming *delta*, not absolute accuracy). A canonicalization front-end would instead add a per-frame eigendecomposition to the hot path and re-open the invariance question after quantization; here the guarantee is a fixed property of the read-out and survives both int8 and causal streaming without re-certification.

6. Conclusion

On the metric the field reports, no architecture separates from another: on clean frontal KIMORE our model, the baseline, and a hand-crafted-invariant GRU tie within a 0.33 MAD nondeterminism floor — the metric is saturated. What separates them is off the frontal manifold. Our read-out is $SE(3)$-invariant by construction and certified to machine roundoff, so relocating the camera moves a patient's score by 9×10^{-6} MAD for arbitrary weights, while a broken tracking node — fatal to the hand-crafted model — is absorbed by the graph encoder; both properties replicate on a second corpus, and the guarantee survives to int8 and to causal streaming. We recommend the field report a mean-predictor floor and the per-sequence degradation *distribution* — not only its mean, which on our own augmented baseline understates what the worst patient experiences by nearly 7× — alongside aggregate error, and evaluate off the frontal manifold — camera relocation and node failure are the ordinary conditions of a home device, not adversarial constructions.

Code availability. KIMORE [1], NTU RGB+D and REHAB24-6 are publicly available from their authors. The equivariance certificate (gates E1–E8), the corruption pipeline, the spectral census, the node-failure harness, the quantization and streaming benchmarks, and the aggregation script that regenerates every number in this paper from banked artifacts will be released publicly on acceptance. The repository URL is withheld here to preserve double-blind review; an anonymized copy is available to reviewers on request through the program chairs.

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